Supplement V.C–V.D: Isotropy–Deformability and the Affine Clifford Lock

EABC Reynolds–Clifford Audit, Version 30

Supplementary mathematical note

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Abstract

We isolate an exact algebraic relation linking the reversible three-edge EABC Markov kernel to the inner particle–hole symmetric deformation mode of its H_3-Reynolds-projected fermionic second quantization on $\mathcal{F} = \Lambda^\bullet \mathbb{C}^3$. The result is internal to the finite-dimensional operator model: the inner Clifford shape contribution vanishes exactly on an explicit quadratic hypersurface in (u, v, w), equivalently in the Markov coordinates $(s, X_{\mathrm{M}}, Y_{\mathrm{M}})$. In the parity-symmetric sector $u = v$, the nontrivial branch reduces to $u = v = 2w$, which is also the weighted two-mode isotropy point $X_{\mathrm{M}} = Y_{\mathrm{M}} = 0$. No claim is made here that the finite-dimensional deformation functional is, by itself, a Yang–Mills mass gap or a continuum particle mass.

1 Setup

Let

$$T(u,v,w) = \begin{pmatrix} 1-2u-2v & 2u & 2v \\ 4u & 1-4u-4w & 4w \\ 4v & 4w & 1-4v-4w \end{pmatrix},$$

with stationary weights $\varpi = (1/2, 1/4, 1/4)$. Put $L = I - T$ and

$$\widehat{L} = \Pi^{1/2} L \Pi^{-1/2}, \qquad \Pi = \mathrm{diag}(1/2, 1/4, 1/4).$$

A direct calculation gives

$$\widehat{L} = \begin{pmatrix} 2(u+v) & -2\sqrt{2}\,u & -2\sqrt{2}\,v \\ -2\sqrt{2}\,u & 4(u+w) & -4w \\ -2\sqrt{2}\,v & -4w & 4(v+w) \end{pmatrix}.$$

On $\mathcal{F} = \Lambda^\bullet \mathbb{C}^3$, let $\Delta_F = d\Gamma(\widehat{L})$. For the $H_3 \cong C_2$ parity generator $R_3 = (-1)^{n_3}$ define the Reynolds projection

$$\Delta_{H_3} = \frac{1}{2}(\Delta_F + R_3 \Delta_F R_3).$$

At one-particle level this is

$$\widehat{L}_{H_3} = \begin{pmatrix} 2u+2v & -2\sqrt{2}\,u & 0 \\ -2\sqrt{2}\,u & 4u+4w & 0 \\ 0 & 0 & 4v+4w \end{pmatrix}. \tag{1}$$

Denote the ordered eigenvalues of the 2×2 block in (1) by $\alpha \le \beta$, put

$$\gamma = 4(v+w), \qquad \kappa = \mathrm{Tr}\,\widehat{L}_{H_3} = 6u + 6v + 8w.$$

The fermionic particle–hole operator C obeys

$$C\Delta_{H_3}C^{-1} = \kappa I - \Delta_{H_3},$$

so the inner Clifford pair is $\{\beta, \kappa - \beta\}$. For a strictly positive pair $\lambda_- \le \lambda_+$ define

$$\eta = \frac{1}{2}\log\frac{\lambda_+}{\lambda_-}, \qquad L_{\text{shape}} = \log\cosh(\pi\eta).$$

Hence the inner contribution vanishes exactly when the two members of the inner pair coincide.

2 The Isotropy–Deformability Equivalence

Theorem 2.1 (Isotropy–Deformability Equivalence)**.** *Assume the inner Clifford pair is positive. Then*

$$L_{1,\text{inner}} = 0$$

if and only if

$$9u^2 - 2uv + 4uw - 3v^2 - 12vw = 0. \tag{2}$$

Equivalently, with

$$s = 3u + 3v + 4w, \qquad X_{\text{M}} = u + v - 4w, \qquad Y_{\text{M}} = 2\sqrt{2}\,(u - v),$$

the vanishing condition is

$$X_{\text{M}}^2 + sX_{\text{M}} + Y_{\text{M}}^2 + 2\sqrt{2}\,sY_{\text{M}} = 0. \tag{3}$$

Proof. The discriminant of the 2×2 block in (1) is

$$d = 2\sqrt{8u^2 + (u - v + 2w)^2}.$$

Since $\kappa = \alpha + \beta + \gamma$, the inner particle–hole pair is degenerate precisely when

$$\beta = \kappa - \beta,$$

which is equivalent to

$$2\beta - \kappa = 0.$$

Using $\beta = (\alpha + \beta + d)/2$ and $\alpha + \beta = \kappa - \gamma$, one obtains

$$2\beta - \kappa = d - \gamma.$$

Therefore $L_{1,\text{inner}} = 0$ if and only if $d = \gamma$. Under the stated positivity assumptions both sides are nonnegative, so squaring introduces no additional branch. Expanding gives

$$d^2 - \gamma^2 = 4\big(9u^2 - 2uv + 4uw - 3v^2 - 12vw\big),$$

which proves (2). Substituting

$$u = \frac{s + X_{\text{M}} + \sqrt{2}Y_{\text{M}}}{8}, \qquad v = \frac{s + X_{\text{M}} - \sqrt{2}Y_{\text{M}}}{8}, \qquad w = \frac{s - 3X_{\text{M}}}{16}$$

reduces the same polynomial exactly to (3). □

Corollary 2.2 (Parity-symmetric branch)**.** *On the parity plane $u = v = t$, the vanishing condition factorizes as*

$$t(t - 2w) = 0.$$

Hence on the nontrivial branch $t > 0$,

$$\boxed{u = v = 2w}.$$

Moreover, on $u = v$ one has $Y_{\mathrm{M}} = 0$, and the condition $u = v = 2w$ is equivalent to

$$X_{\mathrm{M}} = Y_{\mathrm{M}} = 0.$$

Thus, within the parity-symmetric sector, weighted Markov two-mode isotropy and vanishing of the inner Clifford deformation mode coincide exactly.

Remark 2.3 (Scope)**.** *The theorem does* not *state that the full quadratic hypersurface* (2) *is identical to the codimension-two isotropy locus $X_{\mathrm{M}} = Y_{\mathrm{M}} = 0$. The coincidence is exact on the parity plane $u = v$. Away from that plane, $L_{1,\mathrm{inner}} = 0$ defines the larger hypersurface* (3)*.*

3 Empirical finite-shell point

For the audited finite-shell kernel

$$u = 0.1442375, \qquad v = 0.1445125, \qquad w = 0.0660875,$$

one obtains

$$s = 1.1306, \qquad X_{\mathrm{M}} = 0.0244, \qquad Y_{\mathrm{M}} \approx -7.7781746 \times 10^{-4}.$$

The exact-degeneracy polynomial evaluates to

$$F_{\mathrm{deg}} = X_{\mathrm{M}}^2 + sX_{\mathrm{M}} + Y_{\mathrm{M}}^2 + 2\sqrt{2}\, sY_{\mathrm{M}} \approx 2.5695285 \times 10^{-2},$$

and the inner Clifford splitting is

$$\delta_{\mathrm{inner}} \approx 1.5115624 \times 10^{-2}.$$

Consequently

$$\eta_{\mathrm{inner}} \approx 6.6848792 \times 10^{-3}, \qquad L_{1,\mathrm{inner}} \approx 2.2050831 \times 10^{-4} > 0.$$

The parity-averaged ratio is

$$\frac{(u + v)/2}{2w} \approx 1.09230.$$

This last percentage-style distance is a descriptive coordinate diagnostic; it is not a canonical metric distance to the full hypersurface (2).

4 Spectral-Diameter Normalization and Affine Clifford Uniqueness

The particle–hole mirror relation also isolates a canonical centered direction once the admissible inner operator is restricted to the affine spectral class generated by Δ_{H_3}. This restriction is essential: the full real self-adjoint H_3-compatible Clifford-odd commutant is not one-dimensional. In the audited representation, imposing $[B, R_3] = [B, P_0] = [B, P_1] = [B, P_2] = 0$ and $CBC^{-1} = -B$ leaves five real dimensions; adding $[B, \Delta_{H_3}] = 0$ leaves four. One-dimensionality occurs only after the further affine condition $B \in \mathrm{span}_{\mathbb{R}}\{I, \Delta_{H_3}\}$.

Theorem 4.1 (Affine Clifford lock and spectral-diameter normalization)**.** *Let $\Delta = \Delta_{H_3}$ be self-adjoint and satisfy*

$$C\Delta C^{-1} = \kappa I - \Delta, \tag{4}$$

with $\mathrm{spec}(\Delta) \subset [0, \kappa]$ *and both endpoints attained. Consider the affine spectral class*

$$B = \alpha\Delta + \beta I, \qquad \alpha, \beta \in \mathbb{R}.$$

Then the Clifford-odd condition

$$CBC^{-1} = -B$$

is equivalent to

$$B = \alpha\left(\Delta - \frac{\kappa}{2} I\right). \tag{5}$$

If, in addition, the spectral diameter is normalized by

$$\mathrm{spec}(B) \subset \left[-\frac{1}{2}, \frac{1}{2}\right], \qquad \left\{-\frac{1}{2}, \frac{1}{2}\right\} \subset \mathrm{spec}(B),$$

then

$$\boxed{B = \pm B_0, \qquad B_0 := \frac{\Delta - \kappa I/2}{\kappa}.} \tag{6}$$

Thus B_0 is unique up to the global sign within the affine H_3-spectral class.

Proof. Using (4),

$$CBC^{-1} = \alpha(\kappa I - \Delta) + \beta I = -\alpha\Delta + (\alpha\kappa + \beta) I.$$

Equating this to $-B = -\alpha\Delta - \beta I$ gives

$$\alpha\kappa + \beta = -\beta,$$

hence $\beta = -\alpha\kappa/2$ and (5). Because the endpoints 0 and κ occur in $\mathrm{spec}(\Delta)$, the extremal spectral values of B are $\pm|\alpha|\kappa/2$. Requiring these to be $\pm 1/2$ gives $|\alpha| = 1/\kappa$, which yields (6). The sign is not fixed by the spectral-diameter condition. □

Remark 4.2 (Full-commutant limitation)**.** *Theorem 4.1 is deliberately not a uniqueness theorem on the full H_3 commutant. The exact linear audit gives the real dimension cascade*

$$5 \quad \xrightarrow{[B,\Delta_{H_3}]=0} \quad 4 \quad \xrightarrow{B\in\mathrm{span}\{I,\Delta_{H_3}\}} \quad 1.$$

Consequently, the affine spectral-class assumption is a genuine additional structural hypothesis and must not be suppressed when the theorem is quoted.

4.1 Dirac-square consequence inside the affine class

Let $P = \sqrt{\Delta_{\mathrm{rad}}}$ denote the positive radial momentum operator. For the ungraded first-order ansatz

$$D_{\mathrm{aff}} = P \otimes I + I \otimes B_0,$$

one has identically

$$D_{\mathrm{aff}}^2 = P^2 \otimes I + I \otimes B_0^2 + 2P \otimes B_0. \tag{7}$$

Hence the coefficient of the linear radial–internal cross term is fixed to 2 once the normalization (6) has been chosen. This does not imply $B_0^2 = \Delta_{H_3}$; in fact the audited spectrum shows that equality fails. A residual internal potential $\Delta_{H_3} - B_0^2$ is therefore required if the second-order operator is to retain the original H_3 spectrum.

For the audited kernel, the determinant-stationarity calculation with the spectral-diameter normalized B_0 and cross coefficient 2 gives the internal-unit candidate

$$\boxed{M_*^{\text{affine-Clifford}} \approx 1.04923.} \tag{8}$$

Equation (8) is a conditional invariant of the affine-Clifford mixing model. It is not an absolute mass in physical units, nor is it claimed to be universal over the full four-dimensional dynamically compatible commutant.

5 Epistemic status and bridge statement

Statement	Status
Reynolds formula and H_3 block structure	Exact finite-dimensional linear algebra
Particle–hole spectral mirror $C\Delta_{H_3}C^{-1} = \kappa I - \Delta_{H_3}$	Exact for the constructed fermionic lift
Theorem 2.1 / equations (2)–(3)	Exact symbolic identity
Parity corollary $u = v = 2w$	Exact on the nontrivial parity branch
Numerical values at the audited kernel	Finite-shell numerical evaluation
Affine Clifford uniqueness $B = \pm B_0$	Exact within $B \in \text{span}\{I, \Delta_{H_3}\}$ plus spectral-diameter normalization
Full-commutant uniqueness of B_0	False in the audited model; four dimensions remain after $[B, \Delta_{H_3}] = 0$
$M_*^{\text{affine-Clifford}} \approx 1.04923$	Numerical stationary point of the specified affine-Clifford mixing model
Identification with a continuum Yang–Mills mass gap	Not proved
Identification with quark/gluon masses	Not claimed

Bridge Conjecture 5.1 (Bridge to a continuum gap)**.** *Suppose an independently constructed continuum limit carries the finite-dimensional inner Clifford deformation into a self-adjoint continuum operator whose lowest nonzero spectral value is controlled from below by a coercive functional of the distance to the degeneracy hypersurface* (2)*. Then a uniform positive separation from that hypersurface may provide a route to a positive continuum spectral gap. Establishing such a coercive bridge, its scaling, and its renormalization behavior is a separate open problem.*

6 Reproducibility note

The accompanying audit files record: the H_3 particle–hole mirror test, the Clifford-fixed shape summary, the exact-degeneracy gate, the successive commutant-rigidity audit, and the Dirac-square normalization sensitivity. The symbolic certificate uses only the formulas displayed above; no target angle, mass, or coupling constant is injected.